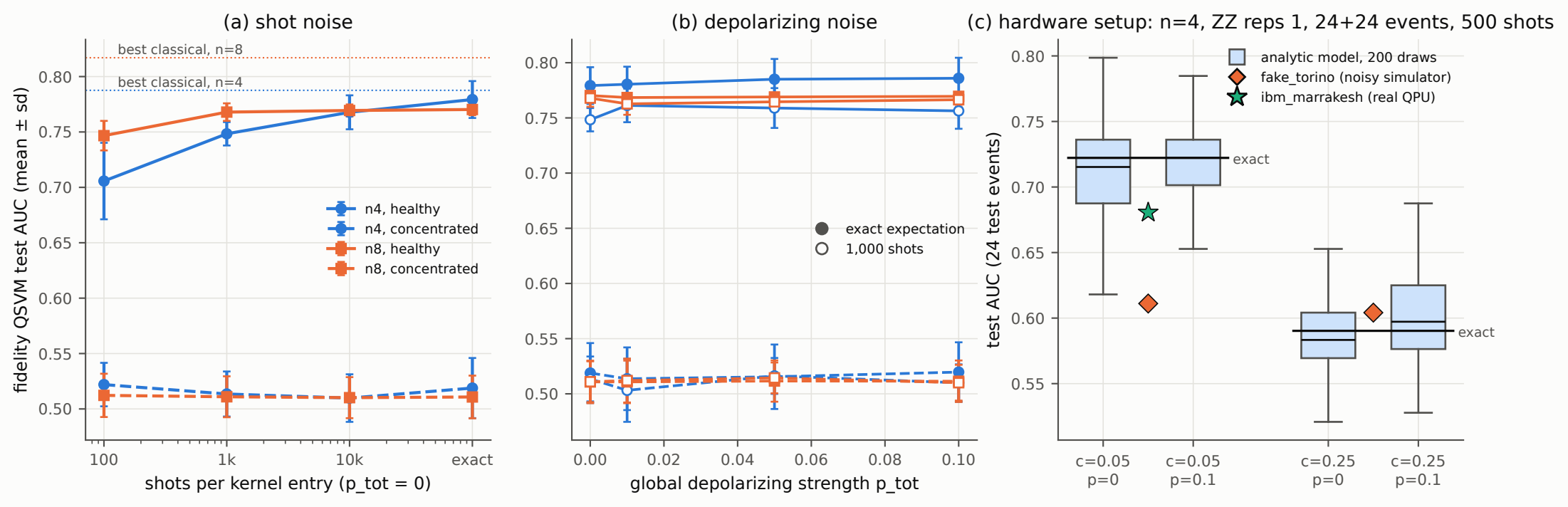
(a) shot noise
best classical, n=8
best classical, n=4
n4, healthy
n4, concentrated
n8, healthy
n8, concentrated
fidelity QSVM test AUC (mean ± sd)
0.80
0.75
0.70
0.65
0.60
0.55
0.50
100
1k
10k
exact
shots per kernel entry (p_tot = 0)
(b) depolarizing noise
exact expectation
1,000 shots
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.00
0.02
0.04
0.06
0.08
0.10
global depolarizing strength p_tot
(c) hardware setup: n=4, ZZ reps 1, 24+24 events, 500 shots
analytic model, 200 draws
fake_torino (noisy simulator)
ibm_marrakesh (real QPU)
test AUC (24 test events)
0.80
0.75
0.70
0.65
0.60
0.55
exact
exact
c=0.05
p=0
c=0.05
p=0.1
c=0.25
p=0
c=0.25
p=0.1